

TIM 80C, Lecture # 10, (4/30/26)

Agenda

1. HW # 3
2. Review of core concepts : FAST, Function Structure, Conceptual Design Process
3. Overview of Financial Strategy: Part 2
4. Fundamental Financial Concepts
 - Present Value (PV), Net Present Value (NPV)
- (?) 5. Cash Flow Analysis for new product development in a start-up

1. HW #3 : A mobile robot start-up

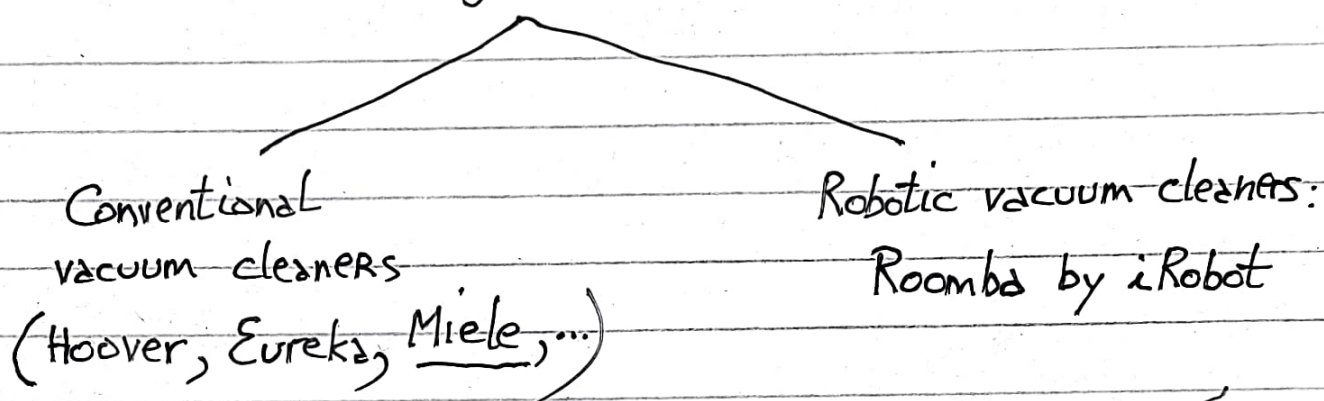
Prob 1. Conceptual Product Design of a robotic vacuum cleaner

Articulate & then implement the 6-step
(Write down)

process : see Product Conceptual design
process on Canvas

Hints & Suggestions :

Dissect existing products



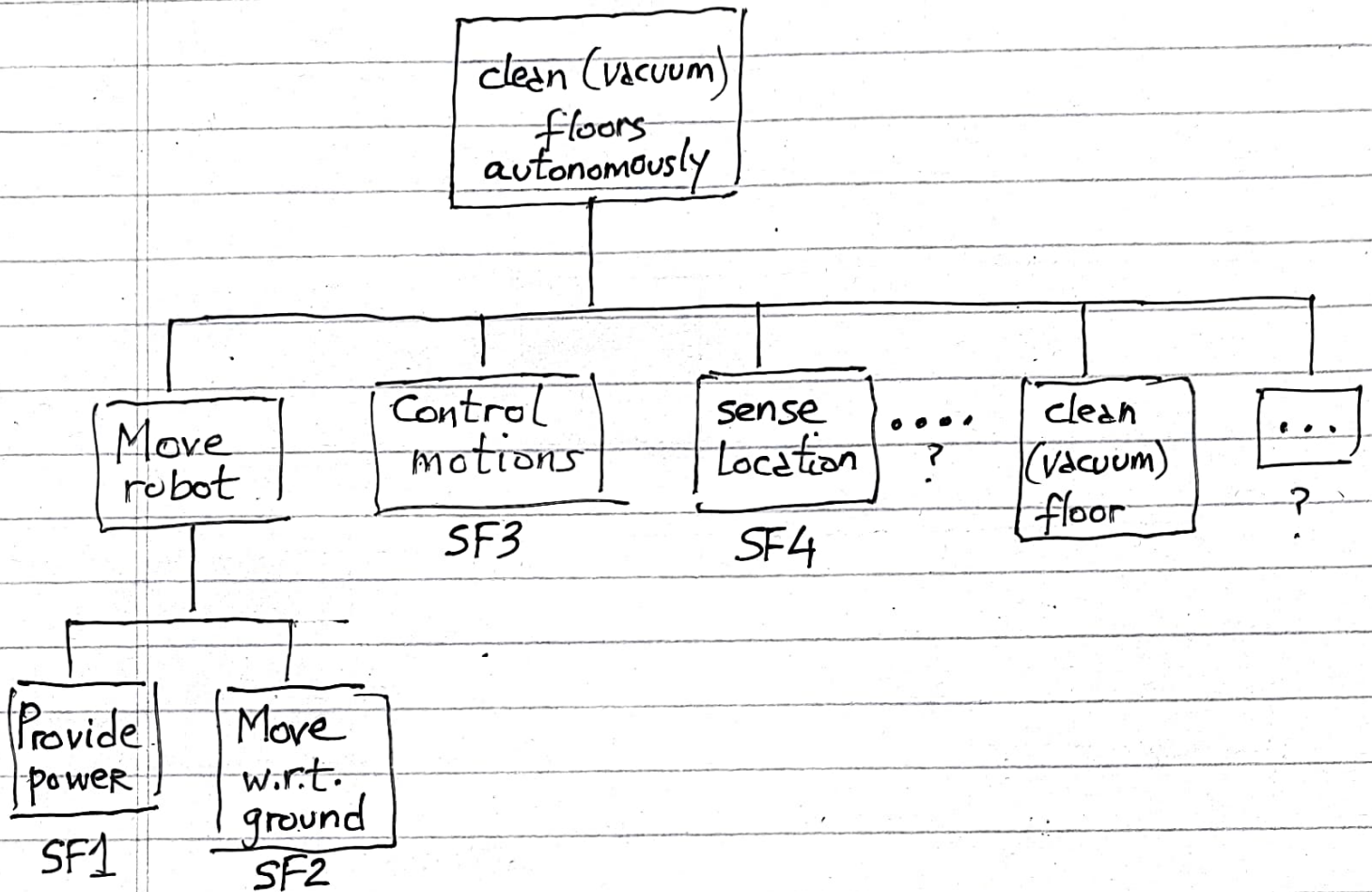
↓
Create FAST diagrams

↓
extract primary function
& sub-functions

↓
create the Function Structure (FS)

⇒ Step 2 of the Product Conceptual Design Process

FS for mobile robot [step 2 of the product conceptual design process]



Step 3 : Morphological Matrix (MM)

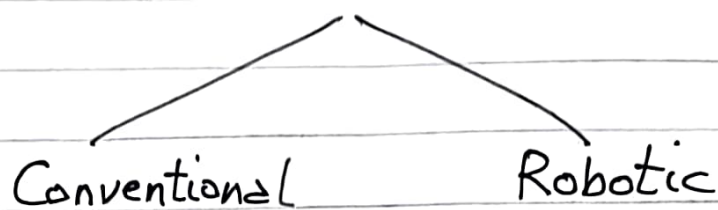
<u>whys</u>	SF1		
	SF2		
	...		
	SF _n		

information
from
Roomba

information
from
another
dissected
product

Step 4 : "Zig-Zag" through the MM

HW#3, Prob 2: Market Strategy for the robotic vacuum cleaner



(Reference: Lecture Notes & The Business Plan Handout, Page 4)

Hints & Suggestions:

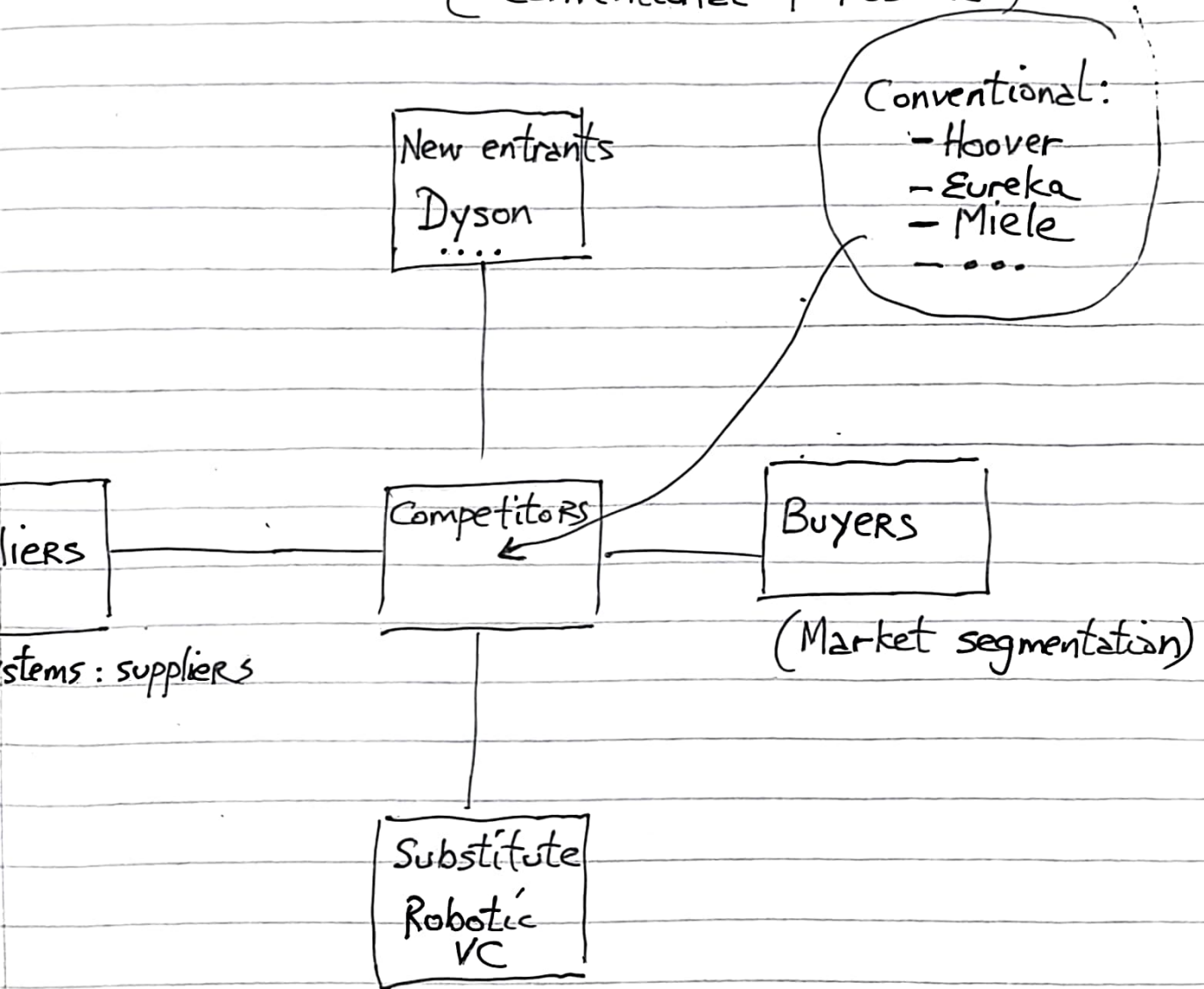
1. Two distinct markets, conventional & robotic
2. Determine the appropriate customer & product segmentation for the vacuum cleaner market.
(See Lecture #8)
3. Create a process for intelligently searching for the data you need: the total size (revenue, \$) of the market; the size (\$) of each market segment, current players, ...
(step 1, 2, 3 of Marketing Strategy)
4. You will not find the most recent information
[The Internet is NOT a data-base]
⇒ Therefore, try to get as close as you can to what is needed.

Prob. 3 : Business / Competitive Strategy

— 3 core steps

Hints :

Entire vacuum cleaner market
(Conventional & Robotic)



3. Overview of Financial Strategy : Part 2

(i) Understand & quantify all relevant cash-flows :

Cash in-flows : Sales Revenue

Cash out-flows : costs related to
product development,
manufacturing, marketing, ...

$$\text{Profit} = \text{"Cash in-flows"} - \text{"Cash out-flows"}$$

(ii) Create income statements, balance sheets,
& cash-flow statements

(iii) Create a financial / funding strategy for
the start-up.